

24/08/26

CSAD702

Model lab Exam

Name:- P. Greethma

Regno:- 192424418

slot :- B

Course:- Computer Network

1. Data Link Layer Traffic simulation using packet Tracer - Analysis of ARP

Aim:- To simulate Data link layer traffic using cisco packet Tracer and Analyze the working of ARP between two devices.

Algorithm:-

1. Open Cisco packet Tracer.
2. place two pcs and one switch
3. connect both pcs to the switch
4. Assign IP addresses to the pcs in the same network
5. switch the simulation mode.
6. Display the ARP Table and verify the learned MAC addresses

Input:-

Pc0 IP addresses : 192.168.1.1

Pc1 IP addresses :- 192.168.1.2

Subnet mask : 255.255.255.0

ping destination: 192.168.1.2

Output:-

ARP Request:- Who has 192.168.1.2 ?

ARP Reply:- 192.168.1.2 is at <MAC Address>

Ping successful.

Result:-

The Data Link layer traffic was successfully simulated in cisco packet Tracer.

2. Implementing sliding window protocol in c

Aim:- To implement the sliding window protocol and demonstrate reliable transmission of multiple frames.

Algorithm:-

1. Start the program
2. Read the total number of frames.
3. Read the window size.
4. Divide the frames into transmission windows
5. Send all the frames within the current flow.
6. If a frame is lost or an acknowledgment is not received, retransmit the required frames.

Input:-

Enter number of frames: 8

Enter window size: 3

Output:-

sending frames: 1 2 3

o/p Acknowledgement received from frame 1
Acknowledgement received from frame 2
Acknowledgement received for frame 3.

All frames transmitted successfully.

Program:-

```
#include <stdio.h>
```

```
int main()
```

```
{  
    sliding window-protocol()
```

```

int n, w, i, j;
printf ("Enter number of frames");
scanf ("%d", &n);
printf ("Enter window size:");
scanf ("%d", &w);
for (i = 1; i <= n; i += w) {
    printf ("\nsending frames:");
    for (j = 1; j < i + w & j <= n; j++)
        printf ("%d", j);
    printf ("\nAcknowledgement received\n");
}
printf ("All frames transmitted successfully.\n");
return 0;
}

```

Result:-

The implementation of sliding window in protocol in C programming successfully generated.